Riley Matthiessen
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Impact of European Green Crabs on the Feeding Behavior and Physiology of Hairy Shore Crabs Over Increasing Water Temperatures

Abstract

European green crabs (*Carcinus maenas*) are an invasive species originally native to western Europe. Now with a potential range all over the world European green crabs are one of the most invasive marine organisms. European green crabs are prolific predators that prey on native species. European green crabs display a large range of thermal tolerance which allows them to continue to be effective invaders in warming sea temperatures. As global ocean temperatures continue to rise, increased green crab presence paired with increased thermal stress will likely impact both the behavior and physiology of native species impacting ecosystem interactions and health. In this study we look into how European green crab presence and temperature increase impact the feeding behavior and metabolism of Washington native hairy shore crabs (*Hemigraspus oregonensis*). Over the course of the study we found possible negative relationships between European green crab presence with time spent feeding and amount of food eaten by hairy shore crabs. This relationship could suggest increased European green crab presence negatively affecting hairy shore crab populations through non-trophic interactions. This project is an initial pilot study aimed to provide a base for further research and suggest possible relationships. Further research into this subject could provide insight into how increasing ocean temperatures will impact native species in green crab ranges and influence management of conservation and crab fisheries operations.

Intro

Introduced to the Americas by European merchants in the 1800s (NOAA, 2026) , invasive European green crabs (*Carcinus maenas*) now pose a large threat to intertidal ecosystems along the East and West coasts of the united states as well as a possible range spanning all continents outside of Antarctica. Multiple locations along the Washington coast were found to be highly suitable to European green crabs and expected to be green crab hotspots in a model evaluation (Harney, J, 2008) which suggested that locations having high wave protection, sand or mud flats,eel grass, and supratidal vegetation were the most suitable habitats for European green crabs to have increased invasion success. Notably, European green crabs are known to have a large range of thermal tolerance (Leignel, et al, 2014) acclimating quickly to a wide range of temperatures making them effective invaders. European green crabs are also predators of many native species along the Washington coast such as hairy shore crabs (*Hemigraspus oregonensis*). Aside from the known predator prey interactions there is limited research into non-trophic interactions between European green crabs and hairy shore crabs or many other native species. As global ocean temperatures continue to rise, increased thermal stress on native species could compound with negative trophic and non-trophic interactions with European green crabs, which are minimally impacted by increased temperatures, leading to significant increase to negative effects on Washington intertidal ecosystems as a result of green crab invasion. During this pilot study, we used a 2x2 experimental setup to test for relationships between European green crap presence on the feeding behaviors and physiology of hairy shore crabs as well as any influence on these relationships by increased temperatures. The results of this study suggest a possible negative relationship between green crab presence and shore crab feeding behavior. This relationship could be important in influencing conservation efforts as decreased feeding in larval shore crabs has been shown to increase mortality(Sulkin et al, 1998). We hope that this study can be used to guide future research to further define the relationships between green crab presence, temperature and feeding behavior, and inform future management decisions regarding conservation and fisheries.

Methods

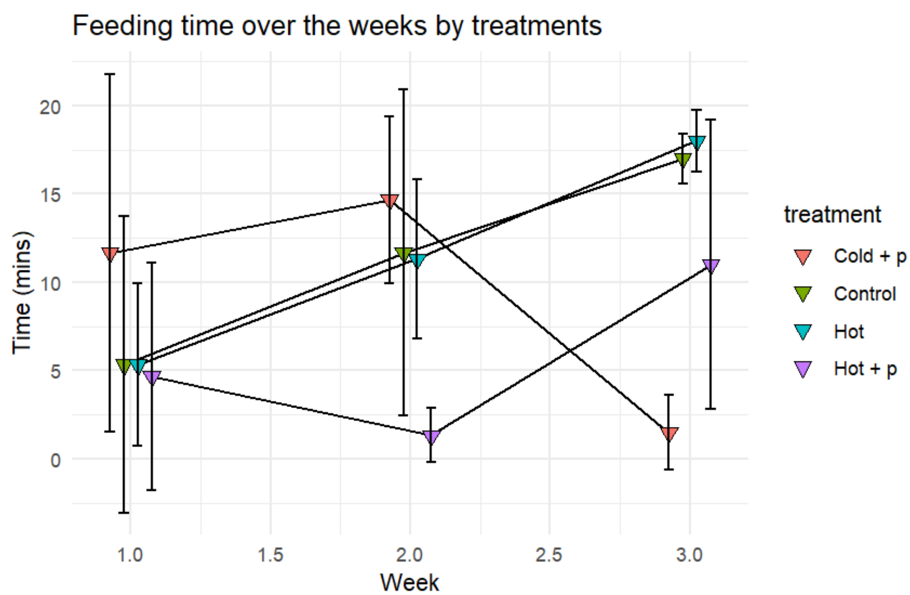
Four separate experimental conditions were prepared in a 2x2 experimental design setup. The conditions consisted of tank one, our control treatment held at typical water temperatures for the Salish Sea of around 14C and without the presence of a European green crab. Tank two was held at typical Salish sea temperatures and held a European green crab. Tank three was held at elevated temperatures at 24C and did not hold a European green crab. Tank four was also held at elevated temperature and did hold a European green crab. All treatments were held at a salinity of 35 ppt. All tanks were prepared with a layer of sandy substrate and a small shell for the shore crabs to hide under. Tanks holding the European green crab were prepared with a plastic mesh divider separating European green crabs and shore crabs. Three hairy shore crabs were selected for each treatment, chosen for relatively uniform size and were tagged with a marker giving each shore crab a unique ID number and placed into their respective tanks. Over the course of the three week experiment, shore crabs in each treatment were fed three small pieces of fish of around 2 grams in weight. Food weight was recorded and was then kept in tanks for 20 minutes per feeding before being removed, dried and weighed again. The time each crab spent feeding over the 20 minutes was recorded. Before the start of each feeding any mortalities were replaced with new shore crabs each given their own new ID number.During week two of the experiment, the decision was made to remove European green crabs from the predator treatments between feedings to avoid further mortality of the hairy shore crabs. After the weekly feeding regiment, shore crab metabolic rate was measured by proxy using a Resazurin protocol. At the end of week three hemolymph was extracted from all shore crabs and analyzed using a total protein assay. The significance of difference in collected data between tanks was measured using a two way anova test using the standard value of 0.05 as the significance threshold.

Results

Impacts of temperature and predator presence on shore crab feeding time:

During week one control and hot treatments had the same average feeding time of 5.33 minutes.The hot with predator treatment had a slightly lower feeding time of 4.66 minutes.The cold with predator treatment had a higher average feeding time of 11.66 minutes. Across the three weeks with both control and hot treatments saw similar growth with average week three feeding times of 17 and 18 minutes respectively. The difference in average feeding time between the two treatments was not statistically different over the entire treatment. During week two, the cold with predator treatment saw a slight increase to 14.66 minutes and the hot with predator treatment saw a decrease to 1.33 minutes. During week three, cold and hot predator treatments showed a decrease to 1.5 minutes and an increase to 11 minutes respectively.

Figure 1:



Graph showing the average time spent feeding in minutes by each treatment over the three week period. Error bars represent standard deviation for each average.

Impacts of temperature and predator presence on shore crab food consumption:

Over the course of the study, the hot with predator treatment group remained statistically similar when it came to change in food mass over time. The cold with predator group showed an increase in average change in food mass from week one to week two followed by a significant decrease in week three.The control group decreased in week two before increasing again in week three. The hot treatment showed an increase in average change in food mass over each week of the study.

Figure 2:

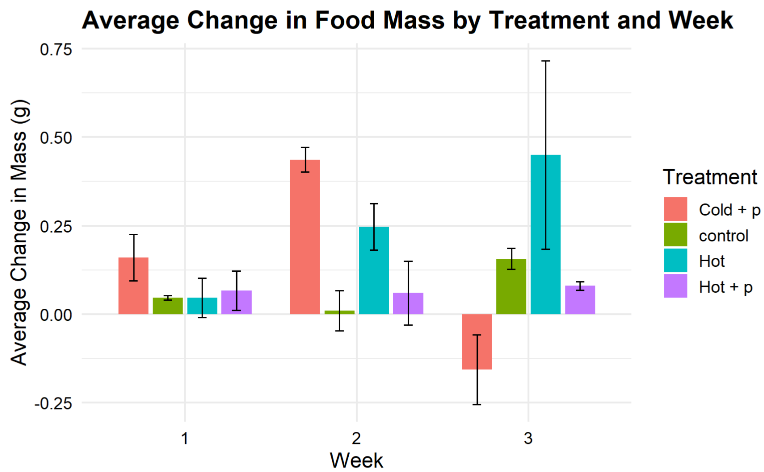


Figure showing the average change in food mass from before to after feeding among crabs in each treatment. Error bars show standard deviation and treatment type is designated by color.

Impacts of temperature and predator presence on shore crab metabolic rate:

All crab treatment groups displayed a significant decrease in average metabolic rate from week one to week three. From week one to week two, the cold with predator, control and hot treatments all displayed a slight but not significant decrease in average metabolic rate. The hot with predator treatment group however displayed an increase in average metabolic rate from weeks one to two. During week one, the control group displayed a significantly higher metabolic rate than the other treatment groups that were all not significantly different.

Figure 3:

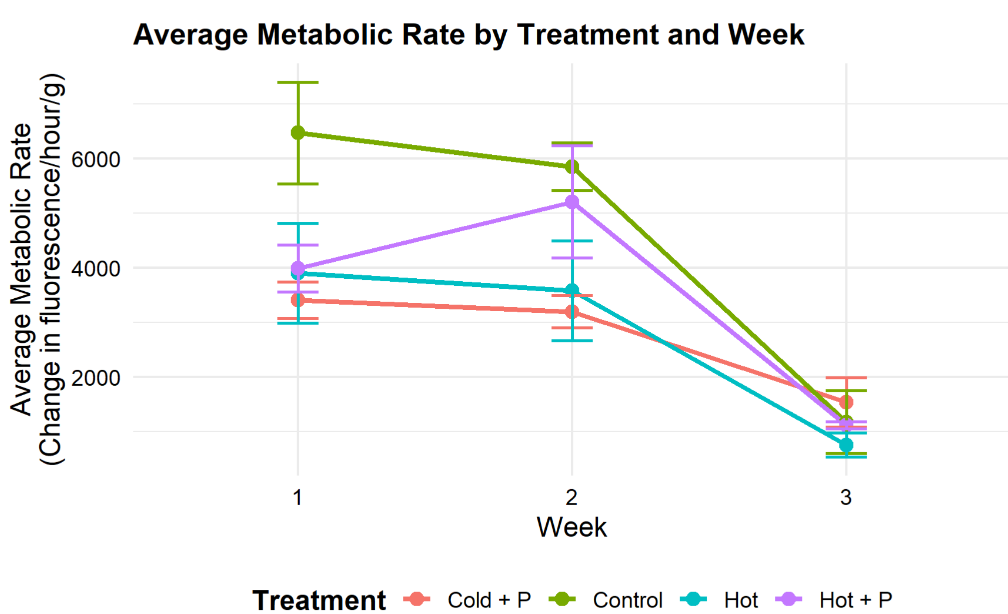


Figure showing change in average metabolic rate among each treatment group. Error bars show standard deviation and treatment type is designated by color.

Discussion

First of all it is worth noting that due to the nature of this project as a pilot study, its low sample size and complications during the experimental process, the data collected has relatively low significance and it is difficult to draw solid conclusions from the observed data. Hopefully however, the results of this project may be used to suggest possibilities for the physiological responses to predator cues in hairy shore crabs, provide a useful starting place for further research or aid in producing more conclusive data in future studies. With the acknowledgment of the complications and shortcomings of the data, we did observe certain patterns in the data that suggest physiological and behavioral responses to changes in temperature and the presence of European green crabs.

Average feeding time data suggests a relationship between predator presence and shore crab feeding behavior. The control and hot, non predator treatments displayed an increase in feeding time over the course of the study not observed in the predator treatments. Both of the non predator treatments had similar linear increases in average feeding time where the predator groups showed fluctuations. This pattern suggests some relation between the absence of predator cues and increased time feeding. This could be due to the absence of stress from predator cues and decreased time spent hiding resulting in crabs spending more time in the open and more time feeding during the feeding regimen.

Coinciding with average feeding time patterns across the study, change in food mass, or the amount eaten by crabs during feeding showed a similar pattern in the hot treatment increasing each week. The decrease in average feeding time in the cold with predators was similar to the decrease in change in food mass observed during week three in the same treatment. High variability and deviation among change in average food mass makes suggesting any relationships between either temperature or predator presence difficult. As change in food mass does not directly correlate with change in feeding time among all treatments it is likely that temperature and predator presence and their resulting behavioral and physiological impacts are forcing change in feeding mass in different directions leading to the high variability and lack of obvious relations observed in this data set.

All treatment groups experienced a similar decrease in average metabolic rate over the course of the three week study period. The hot with predator treatment group displayed an increased metabolic rate from week one to week two however this increase is likely influenced by the introduction of new shore crabs to this treatment type to replace crabs lost to mortality in week two bringing up the average. Interestingly, the hot with predator outlier not included, all other treatments displayed a smaller decrease in average metabolic rate from week one to week two compared to a much greater decrease in average metabolic rate from week two to week three. As this decrease in metabolic rate is observed in all treatment types, this decrease is likely due to an uncontrolled variable. One possible cause for this decrease could be due to the limited feeding schedule of 20 minutes once a week. As some crabs did not eat during every feeding opportunity, a decreased amount of food from their pre-experimental conditions could lead to a decrease in metabolic rate over the study.

Based on these initial patterns, it is likely that both temperature and the presence of predator cues do have an influence on shore crab feeding behavior and physiology. These influences could be further explored in future research to provide more conclusive evidence. A full scale research project conducted to look further into this relation would consist of a much larger sample size, increasing the significance of the data. To further investigate the influences of temperature another temperature treatment could be added as an intermediate between the control and hot treatments used in this study as a 2x3 experimental design. Further research could provide evidence for or against relationships that were unclear from data collected during this study and clarify the extent of the relationships suggested from our data.

References

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